Without blockchain, every company needs to maintain a different database. Blockchain employs a distributed ledger, which ensures that transactions and data are recorded consistently across all locations. Full transparency is offered since any network user with permissions may see the same data at once. All transactions are time- and date-stamped records with immutability. Members may access the whole transaction history thanks to this, which almost eliminates the possibility of fraud.

Blockchain establishes an audit trail that records an asset's origins at each stage of its travel. This helps industries plagued by fraud and counterfeiting. Blockchain makes it feasible to directly communicate provenance information to customers. Traceability data can reveal weak points in any supply chain, such as items stored on a loading dock while being transported.

Traditional paper-intensive processes take a long time, are subject to human mistakes, and frequently need third-party mediation. Transactions may be finished more quickly and effectively by automating these operations with blockchain. The blockchain can hold both documentation and transaction information, doing away with the necessity for paper exchange. Clearing and settlement happens more quickly because there is no need to reconcile several ledgers.

With smart contracts, transactions can be automated, enhancing productivity and accelerating processes even further. The subsequent stage in a transaction or workflow is automatically initiated after pre-specified requirements are met. Smart contracts lessen the need for human involvement and rely less on outside parties to confirm that a contract's provisions have been adhered to. For instance, when a consumer files a claim for insurance, it can be immediately settled and paid once he or she has submitted all the required evidence.

Table 1: Frameworks, platforms and toolkits for blockchain development

Platform and software toolkit	How it can be used in blockchain development
Truffle Suite URL: *https://trufflesuite.com/*	Useful for smart contracts development and blockchain emulation
Solidity URL: *https://soliditylang.org/*	Multi-featured programming language for smart contracts
Vyper URL: *https://vyper.readthedocs.io/*	Secured and contract oriented programming language
Rust URL: *https://www.rust-lang.org/*	Efficient and reliable software for blockchain development
Hardhat URL: *https://hardhat.org/*	Extensible, fast and flexible environment for blockchain development
Embark URL: *https://framework.embarklabs.io/*	Powerful framework for decentralised apps development
ThirdWeb URL: *https://thirdweb.com/*	Framework for Web3 development in the blockchain environment
Geth URL: *https://geth.ethereum.org/*	Implementation patterns and modules for Ethereum

Platforms and toolkits for blockchain development

A number of frameworks and programming platforms are available to develop and deploy blockchain based applications. Table 1 lists some of the important ones.

Blockchain technology has a vast array of development frameworks and platforms for use in many applications. Many of these platforms are open source and free, and can be used to design and develop algorithms in a variety of fields, such as security, authentication, privacy, and cryptocurrency. END

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